An Understanding Kit

Eight results from the corpus, to rebuild by hand

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How to use this. Read one problem's statement. Close the PDF and work on paper until you see *why* it must be true. That means rebuilding it, not verifying it. Hints are in Part II and proofs in Part III, deliberately far away. The problems run from a few minutes to an hour. None needs anything outside this document. A problem that lights up tells you something; one that doesn't tells you something too.

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Part I
Problems

Shared definitions

Definition 1 (two-valued state)**.** For a family $\mathcal{D}$ of subsets of a set Ω closed under complements and countable disjoint unions, a *two-valued state* is $s : \mathcal{D} \to \{0, 1\}$ with $s(\Omega) = 1$ that is additive on disjoint pairs. It is *σ-additive* if additivity holds for countable disjoint families, and *finitely additive* if only for finite ones. The *Dirac* δ_ω is $s(E) = 1 \iff \omega \in E$.

Definition 2 (language, two tokens)**.** A *language* is a relation $\rho \subseteq A \times A$ on a finite alphabet A, read as a digraph. It is *primitive* if some power of its adjacency matrix is entrywise positive (equivalently: strongly connected, and the gcd of its cycle lengths is 1). *Two tokens* sit on distinct vertices and move *synchronously*: at each step each token crosses one arc of ρ. They may never occupy the same vertex at the same time.

1 Tiny

Problem 1 (stationary two-valued states are fixed points)**.** *Let A be countable and $\Omega = A^{\mathbb{Z}}$ with the product σ-algebra.*

(a) Every σ-additive two-valued state on Ω is a Dirac δ_ω.

(b) If it is also shift-invariant ($s(S^{-1}E) = s(E)$), then ω is a constant sequence.

Why it matters. "Two-valued" means *one trajectory*, not "deterministic dynamics". So a genuine stationary process is never two-valued. This is why the σ-essential lane sits off the path to dynamics.

Source: not written elsewhere; derived 2026-10-09 in conversation.

Problem 2 (out-of-phase tokens never collide)**.** *Let $\lambda : R \to \mathbb{Z}_c$ be a function on a set of vertices. Token 1 is only ever at $w \in R$ at times $t \equiv \lambda(w)$, and token 2 only at times $t \equiv \lambda(w)+e \pmod{c}$, with $e \not\equiv 0$. Show they never collide in R.*

Why it matters. This is the engine of the reach-lemma attack. A phase difference is a collision-proof shield.

Source: `papers/reconstruction/notes/joint2_wielandt_finding.md`, "(2) Out-of-phase tokens never collide".

Problem 3 (the ascending ring)**.** *Fix $L \geq 2$ and let $C_k = \{x \in \mathbb{N}^L : x_k < x_{k+1}\}$, indices mod L.*

(a) No point x lies in every C_k.

(b) Suppose that, for each window $\{k, k+1\}$, s_k is a σ-additive two-valued state on the σ-algebra of events depending only on (x_k, x_{k+1}). Suppose also that adjacent windows agree on events of their shared coordinate, and that $s_k(C_k) = 1$ for all k. Show this is impossible.

Why it matters. Drop "σ-" in (b) and the configuration *exists*: an ordered product ultrafilter makes every window an ascent. That finitely additive object, glued on a carrier, is the Lean-checked countable σ-essential witness (every $L \geq 4$).

Source: `notes/open_questions/delay_embedding/collision_vs_generation.md` (update 2026-10-09); Lean `formalization/staginsL_sigmaEssential`.

2 Small

Problem 4 (the cycle lemma)**.** *Let ρ be primitive and $C = (x_0, \dots, x_{c-1})$ a simple cycle, $c \geq 2$. Two tokens* chase *on C at offset $j \neq 0$: both follow C, token 2 being j positions ahead. Show that:*

(a) from any offset, by legal non-colliding moves and back again, the tokens can reach a chase at any other offset;

(b) among the closed two-token runs so produced there are lengths with gcd 1.

Why it matters. This is the reason chases are *aperiodic*. Aperiodicity is exactly what the flagship's forcing lemma needs.

Source: `joint2_wielandt_finding.md`, "CYCLE LEMMA (proved 2026-10-09)".

Problem 5 (one chase class)**.** *Let C, C' be simple cycles of length ≥ 2 sharing a vertex v. Find a single synchronous step that turns some chase on C into a chase on C'. Conclude, using that the simple cycles of a strongly connected digraph are chained by shared vertices, that all chases on all cycles are mutually reachable.*

Source: `joint2_wielandt_finding.md`, "LEMMA K*".

Problem 6 (no witness in the Boolean case)**.** *Let $\mathcal{L}$ be a Boolean σ-algebra on Ω and $B \subseteq \mathcal{L}$ finite and closed under complement. Let s_0 be a two-valued assignment on B that extends to* some *finitely additive two-valued state μ on $\mathcal{L}$. Show s_0 extends to a Dirac.*

Why it matters. This is the baseline the whole σ-essential lane pushes against. Coherence supplies μ, and distributivity turns μ into a point. A witness must break the second step.

Source: `papers/sigma_essential/sigma_essential_body.tex`, Prop. `prop:boolean`.

3 Medium

Problem 7 (Dirac test and localization)**.** *With B finite and complement-closed and s_0 two-valued on B, put $K(s_0) = \bigcap\{E \in B : s_0(E) = 1\}$.*

(a) δ_ω extends s_0 iff $\omega \in K(s_0)$.

(b) Hence a pattern that is finitely coherent but has no σ-additive extension needs $K(s_0) = \varnothing$ and *must kill every non-Dirac σ-additive extension. Explain why the second clause is where all the difficulty lives.*

Source: `sigma_essential_body.tex`, Lemma `lem:dirac-iff`, Thm `thm:localization`, Remark `rem:localization-scope`.

Problem 8 (the parity theorem, odd half)**.** *The golden-mean language on $\{0,1\}$ forbids 11. Observe it around a ring of length L through the L adjacent-pair windows. Let each window assign probability $\frac{1}{2}$ to each of 01 and 10.*

(a) The windows agree on shared sites.

(b) For odd L, no probability measure on legal ring configurations has these window marginals.

(c) Stretch. *For even L every coherent datum is realisable. (Hint section only.)*

Why it matters. This is the flagship's smallest real instance of contextuality, and it is determined by parity.

Source: `papers/reconstruction/notes/archive/reconstruction_body_PROVED_ARCHIVE.tex`, § "The parity theorem".

For comparison. Re-read your own CE / Stone route (`papers/paper_i/probability_from_observation_bod` "Stone realization from finitely additive data") and notice what the click felt like.

Part II

Hints

1. (a) For each time k, the sets $\{x_k = a\}$, $a \in A$, are a countable partition. How many can be true? (b) What is $s \circ S^{-1}$ for a Dirac?

2. Compare the two congruences at a hypothetical meeting.

3. (a) Follow the inequalities around. (b) Show each s_k is a Dirac at some pair, then glue.

4. An *excursion* is a walk $x_i \to x_{i'}$ whose interior avoids C. While one token takes it, the other stays on C. Track the offset. Which quantity changes it? Why do these quantities generate $\mathbb{Z}_c$? (Think about closed walks at x_0 and primitivity.) Removing one vertex from a Cayley graph: what keeps it connected?

5. Put token 1 at v and token 2 one step behind v on C.

6. The finite Boolean algebra generated by B has atoms. Exactly one of them is true.

7. (a) Use complements: every false constraint is a true constraint in disguise.

8. (b) If every window is a.s. 01 or 10, what must a global configuration look like? (c) The coherence polytope is the fractional stable-set polytope of the cycle. Look at $\det(I + P)$ for the cyclic shift P.

Part III
Proofs

Problem 1. (a) For each k the events $\{x_k = a\}$ $(a \in A)$ are countably many, disjoint, and cover Ω, so by σ-additivity exactly one, say a_k, has value 1. Let $\omega = (a_k)_k$. Then $\{\omega\} = \bigcap_k \{x_k = a_k\}$, a countable intersection of value-1 events. Its complement is a countable union of value-0 events, which can be disjointified into value-0 events, so it has value 0 and $s(\{\omega\}) = 1$. Hence $s(E) = 1 \iff \omega \in E$. (b) $\delta_\omega \circ S^{-1} = \delta_{S\omega}$. Invariance gives $\delta_{S\omega} = \delta_\omega$, so $S\omega = \omega$, so ω is constant. □

Problem 2. A meeting at w at time t needs $t \equiv \lambda(w)$ and $t \equiv \lambda(w) + e$, so $e \equiv 0$, a contradiction. □

Problem 3. (a) $x_0 < x_1 < \cdots < x_{L-1} < x_0$. Equivalently, take k maximising x_k; then $x_k < x_{k+1}$ contradicts maximality. (b) As in Problem 1(a), each s_k is $\delta_{(a_k, b_k)}$ on its window, since $\mathbb{N}^2$ is countable and the singletons are measurable. Agreement on the shared coordinate gives $b_k = a_{k+1}$. So one point $x = (a_k)$ makes every C_k true, which contradicts (a). □

Problem 4. *Case $c = 2$.* The single offset is the swap $(x_0, x_1) \to (x_1, x_0)$, a closed run of length 1.

Case $c \geq 3$.

(i) *Defect.* An excursion $x_i \to x_{i'}$ of length ℓ has *defect* $\Delta = \ell - ((i' - i) \bmod c)$. Loops and chords count.

(ii) *Moves.* If token 1 takes it while token 2 stays on C, then token 2 advances ℓ while token 1 lands $\ell - \Delta$ further along C. So the offset changes by $+\Delta$, and by $-\Delta$ if token 2 moves. The interior avoids C, so the only possible collision is at the landing. The move is therefore legal iff the new offset is $\neq 0$.

(iii) *The defects generate $\mathbb{Z}_c$.* A closed walk at x_0 splits into arcs of C and excursions, so its length is $\equiv \sum \Delta \pmod c$. By primitivity these lengths have gcd 1, so they generate $\mathbb{Z}_c$.

(iv) *Connectivity, giving (a).* The offset graph is $\mathrm{Cay}(\mathbb{Z}_c, \pm\{\Delta\})$ with the vertex 0 removed. A connected vertex-transitive graph of degree ≥ 2 is 2-connected (Watkins 1970), so it stays connected.

(v) *Lengths, giving (b).* The chase itself is a closed run of length c. For each Δ, start at an offset $\notin \{0, -\Delta\}$. Token 1 takes the excursion, then token 2 takes it when it arrives. That is a closed run of length $\equiv \Delta$. Since the Δ's generate $\mathbb{Z}_c$, the gcd is 1. □

Problem 5. Token 1 is at v and token 2 at v's predecessor on C. In one step token 1 moves to v's successor on C' (which is $\neq v$, since $|C'| \geq 2$) and token 2 moves onto v. Both are now on C', one apart: a chase on C'. The symmetric step goes back. Chains of cycles sharing vertices connect any two cycles, and a loop in a chain can be skipped, because its neighbours share its vertex. With Problem 4(a), everything is one class. □

Problem 6. The finite algebra B_0 generated by B has atoms that partition Ω. By finite additivity exactly one atom a has $\mu(a) = 1$, and $a \neq \varnothing$. Every element of B_0 contains a or is disjoint from it. So for $\omega \in a$, δ_ω agrees with μ, hence with s_0, on B. □

Problem 7. (a) δ_ω extends s_0 iff $\omega \in E$ for every true E, and $\omega \notin E$ for every false E. A false E has true complement in B, so its constraint is already a true one. Hence the condition is just $\omega \in K(s_0)$. (b) σ-additive two-valued states are Dirac or not. (a) handles the Diracs. The empty-kernel condition is easy to arrange (Navara–Pták does it with even a σ-additive global extension). So the whole difficulty is destroying every non-Dirac σ-additive repair while keeping finite coherence. □

Problem 8. (a) Every window gives each site value 1 with probability $\frac{1}{2}$. (b) Almost surely every window is 01 or 10, so a realising configuration alternates $0, 1, 0, 1, \ldots$ around the ring. That needs L even. (c) The coherence polytope is $\{u \in [0,1]^L : u_i + u_{i+1} \leq 1\}$ and R is the stable-set polytope. The constraint matrix is $I + P$, with $\det(I + P) = 1 - (-1)^L$; its proper square submatrices are path-incidence matrices, with determinants in $\{0, \pm 1\}$. For even L the matrix is totally unimodular, so the polytope is integral and $C = R$. Full proof: archive §"The parity theorem". □

Part IV

Appendix: definitions, notation and background results

Everything the problems use, in one place. The table says what to read for each problem. Entries marked *(standard)* are textbook facts, quoted without proof.

Problem	Read
1	A.1–A.4, A.6
2	C.6
3	A.1–A.4, A.7
4	B.1–B.6, C.1–C.5, D.1–D.3
5	B.1–B.4, B.7, C.1–C.3
6	A.1–A.5, E.1–E.3
7	A.1–A.5, E.1–E.5
8	F.1–F.5

A Measure-theoretic basics

A.1 σ-algebra, Dynkin system. A *σ-algebra* on Ω is a family of subsets containing Ω and closed under complements and countable unions. A *Dynkin system* (or σ-class) only requires closure under complements and *countable disjoint* unions. It need not be closed under intersections. That gap is exactly what makes non-Boolean ("quantum-logic-like") carriers possible.

A.2 Product σ-algebra. On $\Omega = A^I$ (sequences indexed by I, e.g. $I = \mathbb{Z}$ or $\{0, \dots, L-1\}$) it is the smallest σ-algebra making each coordinate $x \mapsto x_k$ measurable. It is generated by the *cylinders* $\{x : x_{k_1} = a_1, \dots, x_{k_m} = a_m\}$. For countable A, every singleton $\{\omega\}$ of $A^{\mathbb{Z}}$ is measurable, being a countable intersection of cylinders.

A.3 Two-valued state; finite vs. σ-additivity. See the shared definitions in Part I. For a two-valued s:

- finite additivity means: of two disjoint sets, at most one is true, and their union is true iff one of them is;
- σ-additivity means the same for countable disjoint families.

Consequences:

- $s(E^c) = 1 - s(E)$;
- s is monotone ($E \subseteq F$, $s(E) = 1 \Rightarrow s(F) = 1$);
- in a σ-algebra, a countable intersection of true sets is true when s is σ-additive.

A.4 Dirac. $\delta_\omega(E) = 1 \iff \omega \in E$. It is always σ-additive. "Two-valued σ-additive = Dirac" holds on any σ-algebra in which a countable partition into singletons (or a countable separating family) is available. Problem 1(a) is an instance.

A.5 Boolean algebra, atoms. A *Boolean algebra* of sets is closed under complement, finite union and finite intersection. A *finite* one is generated by finitely many sets, and its *atoms* (minimal nonempty members) partition Ω. Every member is a union of atoms *(standard)*. "Boolean" = distributive = intersection-closed, in the concrete setting.

A.6 Shift, stationarity. The shift $S : A^{\mathbb{Z}} \to A^{\mathbb{Z}}$ is $(Sx)_k = x_{k+1}$. A state s is *stationary* (*shift-invariant*) if $s(S^{-1}E) = s(E)$ for all E. For Diracs, $\delta_\omega \circ S^{-1} = \delta_{S\omega}$. A *process* is a stationary probability measure on $A^{\mathbb{Z}}$.

A.7 Ultrafilter (for the "why it matters" notes). An *ultrafilter* p on $\mathbb{N}$ is a finitely additive two-valued state on all subsets of $\mathbb{N}$. It is *non-principal* if it is not a Dirac, i.e. every finite set is false. Non-principal ultrafilters exist by Zorn's lemma, and they are *never* σ-additive. The *ordered product* $p \otimes p$ makes $\{(a, b) : a < b\}$ true: "the second coordinate is chosen after, and above, the first". This is the device that fakes an ascending ring.

B Digraphs and languages

B.1 Digraph vocabulary.

- A *walk* is a sequence of vertices joined by arcs. Its *length* is the number of arcs.
- A walk is *closed* if it ends where it starts.
- A *simple cycle* is a closed walk with no repeated vertex other than the endpoints. A *loop* is an arc $v \to v$, a cycle of length 1.

B.2 Strongly connected. Every vertex reaches every other by a walk.

B.3 Period, primitive. For a strongly connected digraph, the *period* is the gcd of all closed-walk lengths, equivalently of all simple-cycle lengths. It is *aperiodic* (*primitive*) if the period is 1. Equivalently, some power of the adjacency matrix is entrywise positive, and then all larger powers are too *(standard; Perron–Frobenius / Wielandt).* In words, for large n there is a walk of length exactly n between any two vertices.

B.4 Language. A relation $\rho \subseteq A \times A$, used as the digraph of legal consecutive symbols. "Primitive language" means its digraph is primitive.

B.5 Excursion, chord, defect. Fix a simple cycle $C = (x_0, \ldots, x_{c-1})$.

- An *excursion* is a walk from some x_i to some $x_{i'}$ whose interior vertices are all off C, other than one of C's own arcs.
- A *chord* is an excursion of length 1, i.e. an arc $x_i \to x_{i'}$ that is not C's arc.
- Its *defect* is $\Delta = \ell - ((i' - i) \bmod c)$: how much longer the excursion is than the arc of C it bypasses, mod c.

B.6 Decomposing a closed walk. A closed walk starting and ending on C splits uniquely into arcs of C and excursions, by cutting at every visit to C.

B.7 Chain-connectivity of cycles *(standard; proved in the repo, session 5).* In a strongly connected digraph, any two simple cycles are joined by a chain of simple cycles in which consecutive cycles share a vertex.

C Two tokens

C.1 Pair graph D. Vertices are ordered pairs (a, b) with $a \neq b$ (the two tokens' positions). There is an arc $(a, b) \to (c, d)$ iff $a \to c$ and $b \to d$ are arcs of ρ and $c \neq d$. A walk in D is a synchronous two-token run that never collides.

C.2 Unordered pair graph $\bar{D}$. The same, with (a, b) and (b, a) identified. A closed walk in $\bar{D}$ returns the tokens to the same *set* of positions, possibly swapped. A *swap* (crossed walk) is a run $(a, b) \rightsquigarrow (b, a)$.

C.3 Chase, offset. On a simple cycle C, a *chase* has both tokens on C, following it, with token 2 a fixed number $j \neq 0$ of positions ahead (the *offset*). It is a closed run of length c. A pair of vertices is *cocyclic* if both lie on one simple cycle, and then it is a chase position.

C.4 SCC, recurrent. A *strongly connected component* (SCC) is a maximal set of mutually reachable vertices. A pair is *recurrent* if it lies on a closed walk of D (equivalently, of $\bar{D}$), i.e. it can return to itself.

C.5 Aperiodic SCC; why it matters. An SCC is aperiodic if the gcd of its closed-walk lengths is 1. The flagship's forcing lemma needs every SCC of $\bar{D}$ that carries a swap to be aperiodic. Otherwise its unsafe ring lengths could sit in a residue class like $3\mathbb{Z}$, which would be a "fork".

C.6 Phase. If walks from a start y to w all have the same length mod c, call it $\lambda(w)$. A token starting at y at time 0 is at w only at times $\equiv \lambda(w)$. A second token can be *out of phase* by e: always at w at times $\equiv \lambda(w) + e$.

D Cyclic groups and Cayley graphs

D.1 $\mathbb{Z}_c$, generation. $\mathbb{Z}_c = \{0, \ldots, c-1\}$ under addition mod c. A set S *generates* $\mathbb{Z}_c$ iff $\gcd(S \cup \{c\}) = 1$. If a set of integers has gcd 1, its residues mod c generate $\mathbb{Z}_c$.

D.2 Cayley graph. $\mathrm{Cay}(\mathbb{Z}_c, \pm S)$ has vertices $\mathbb{Z}_c$ and an edge g—$g \pm s$ for each $s \in S$. It is connected iff S generates $\mathbb{Z}_c$, and it is *vertex-transitive*: every vertex looks the same.

D.3 Watkins' theorem *(standard; Watkins 1970).* A connected vertex-transitive graph of degree d has vertex connectivity at least $\frac{2}{3}(d+1)$. In particular, for $d \geq 2$ removing one vertex leaves it connected.

E Quantum-logic side (σ-essential lane)

E.1 Concrete σ-orthoposet. A family $\mathcal{L}$ of subsets of Ω, containing Ω, closed under complement ($E^{\perp} = E^c$) and under countable unions of pairwise disjoint members. This is a Dynkin system (A.1) viewed as a "logic". Disjoint sets play the role of *orthogonal* events.

E.2 $\perp$-closed sub-orthoposet; local pattern. A finite $B \subseteq \mathcal{L}$ closed under complement: the finitely many events actually being reported. A *two-valued state on B* (local pattern) s_0 assigns 0/1 with $s_0(E^c) = 1 - s_0(E)$ (and additivity on disjoint pairs within B).

E.3 Finitely coherent. s_0 extends to some *finitely additive* two-valued state on all of $\mathcal{L}$.

E.4 Kernel. $K(s_0) = \bigcap\{E \in B : s_0(E) = 1\}$, the points compatible with every reported "yes".

E.5 σ-essential contextual state. A local pattern that is finitely coherent but extends to *no* σ-additive two-valued state on $\mathcal{L}$: coherence that only finite additivity can honour. The Navara–Pták example (1983) has empty kernel yet a σ-additive extension, which shows that "empty kernel" alone is not enough.

F Commensurability side (flagship)

F.1 Ring protocol, windows. Sites $\mathbb{Z}_L$ around a ring. Each adjacent pair $\{i, i+1\}$ is a *window*, reported as a probability table over the legal pairs of ρ.

F.2 Coherent (EA) vs. realisable (PR). The window data are *coherent* (empirically adequate) if overlapping windows give the same single-site marginals. They are *realisable* if some probability measure on legal ring configurations has exactly these window marginals. *Contextual* means coherent but not realisable. A protocol is *commensurable* if every coherent datum is realisable.

F.3 Polytopes C, R. Coherent data form a polytope C, and realisable data the convex hull $R \subseteq C$ of the deterministic configurations. Commensurable $\iff C = R$.

F.4 Stable sets. In a graph, a *stable (independent) set* has no two adjacent vertices.

- STAB is the convex hull of their indicator vectors.
- FSTAB $= \{u \in [0,1]^V : u_i + u_j \leq 1 \text{ on edges}\}$ is its fractional relaxation.
- For the golden mean on a ring, $C \cong \mathrm{FSTAB}(C_L)$ and $R \cong \mathrm{STAB}(C_L)$, with $u_i = \Pr[x_i = 1]$.

F.5 Total unimodularity *(standard; Hoffman–Kruskal).* A matrix is *totally unimodular* if every square submatrix has determinant in $\{0, \pm 1\}$. Then $\{u : Mu \leq b,\, u \geq 0\}$ has integral vertices for every integral b. Also used:

- $\det(I + P) = \prod_k (1 + \omega^k) = 1 - (-1)^L$ for the cyclic shift P, whose eigenvalues are the L-th roots of unity ω^k;
- path-incidence matrices have determinants in $\{0, \pm 1\}$ (triangularise along the path).